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21AOE01- Computer Vision

Activity-1 Gray level Transformations

1. Medical Image Enhancement

1) Which gray-level transformation would you recommend

I recommend Contrast stretching to improve the visibility of darker tissue in the X-ray image

2) Why is it suitable?

Contrast stretching increase the intensity range of the image. it Improve the difference between dark and bright regions, making the tissue easier to observe

3) What effect will it have on the Image?

The X-ray image will have better contrast. Bones and darker tissue will become clearer, helping doctor examine the image more easily

2) Night-time surveillance.

1) Which transformation would improve the visibility?

I recommend log transformation because most pixel value are concentrated in the lower intensity range of 0 to 40.

2) Why is it more suitable than Negative transformation?

Log transformation expands low-intensity pixel value and reveals details in dark areas.
Negative transformation reverses dark and bright values but does not specifically enhance dark region in the same way.

3) Predict the appearance of the enhanced image

The parking lot image will appear brighter and objects such as vehicles, people and shadow will become more visible. Details That were difficult to see at night will be easier to identify.

3) Fingerprint Recognition

1) Which gray level transformation would you apply:

I recommend Contrast Stretching to improve the Contrast between the ridges and Valleys of the fingerprint

2) Explain why

Contrast Stretching increases the difference b/w the intensity value of ridges and Valleys. This makes the fingerprint patterns clearer and helps the recognition system identify unique fingerprint features

3) What would happen if a log transformation were applied instead?

Log transformation would brighten darker regions and reveal some hidden details. However, it may not improve the contrast between ridges and valleys as effectively as contrast stretching, depending on the original image.